Optimal Confidence Procedures for Partially Identified Parameters

Ian Pitman

November 1, 2026

Latest Version Here

Abstract

I construct optimal confidence procedures in a flexible class of partially identified models. Under my decision-theoretic framework, reporting a confidence set incurs a loss equal to its excess length: the length of the part that does not overlap with the identified set. A confidence procedure is optimal if it minimizes worst-case expected excess length. I show that the problem of constructing such a procedure reduces to solving a single optimal testing problem and inverting the resulting test. This theoretical insight allows me to leverage recent computational advances in optimal testing to numerically approximate the optimal procedure. In a series of calibrated simulations, I quantify the suboptimality of existing inference techniques across a variety of empirical settings. Unlike the optimal procedure, methods with poor performance often direct power towards alternatives that are inconsistent with the model. I apply the optimal procedure to a recent event study design and find that it yields confidence sets that are 15 to 22 percent shorter than existing approaches.